Say it, then write it: choose an equation

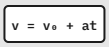
Language goal: I can **explain** why I chose an equation.



given
dato · dado
a number the problem tells you



unknown
incógnita · incógnita
the value you must find



equation
ecuación · equação
a rule that links the values



constant acceleration
aceleración constante · aceleração constante
v changes the same amount each second

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: given · unknown · equation · time · steady

First I list each ____ value. Then I name the ____ I must find. I choose the ____ that uses only what I have. If no ____ is given, I use $v^2 = v_0^2 + 2a\Delta x$. These equations work only when acceleration is ____.

2 · SAY IT TO A PARTNER, THEN WRITE IT



I know ____, ____, and ____. I need ____ .
I choose ____ because it does not need ____ .

3 · SOLVE WITH THE STEPS



0 → ? m/s
a = 3 m/s² · 4 s

A car starts at 0 m/s and speeds up at 3 m/s² for 4 s. How fast is it going?

STEP 1 · LIST
v_0 = ____ · a = ____ · t = ____

STEP 2 · CHOOSE + SOLVE
$v = v_0 + at$ = ____ + ____ × ____

STEP 3 · SAY IT
"After 4 s the car moves ____ m/s."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ I chose this equation because ____ .
☐ I did not use ____ because the problem does not give ____ .
☐ First I listed ____, then I found ____ .

I said it ☐ · I wrote it ☐

R5 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unlock the code: constant acceleration

08 C

DIRECTIONS · Solve each station. Copy the letter into the lock.

STATION 1

v_0 = 1 m/s, a = 2 m/s², t = 2 s. Final v?

(S) 5 m/s (T) 4 m/s (A) 3 m/s

STATION 2

From rest, a = 3 m/s² for 4 s. $\Delta x = \frac{1}{2}at^2$?

(P) 24 m (E) 12 m (K) 48 m

STATION 3

0 → 8 m/s in 4 s, steady. Δx using average v?

(E) 16 m (O) 32 m

STATION 4

20 m/s, a = −4 m/s², to a stop. Δx ($v^2 = v_0^2 + 2a\Delta x$)?

(E) 50 m (I) 100 m (U) 5 m

STATION 5

Steady +2 m/s for 3 s. Δx?

(D) 6 m (N) 5 m

STATION 6

Which equation needs NO time?

(S) $v^2 = v_0^2 + 2a\Delta x$ (R) $v = v_0 + at$

LOCK 1 2 3 4 5 6

Hint: a word for velocity × time… or what a car does at a green light.

R5 1 2 3

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ANTOCCI ENGINEERING · DWG U1-08 · R5

BRAKE ZONE: SCHOOL CROSSWALK

Design spec: cars approach at 15 m/s. The crosswalk is 25 m past the warning sign.

CALC 1 · NEEDED BRAKING

Stop in 25 m: use $v^2 = v_0^2 + 2a\Delta x$. $a =$ ____ m/s²

CALC 2 · STOPPING TIME

$v = v_0 + at \rightarrow t =$ ____ s

DRAWING · SKETCH THE v–t GRAPH FOR THE STOP (label the area)

CALC 3 · CHECK WITH AREA

$\Delta x = \frac{1}{2}(v_0 + v)t =$ ____ m. Does it match 25 m?

CALC 4 · WET ROAD

If the best braking is only −3 m/s², how far is the stop? Safe?

ENGINEER'S NOTE TO THE CITY

Where should the warning sign go? Justify with your numbers.

R5 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Four steps, every equation

08 M

DIRECTIONS · Work each problem through all four boxes.

PROBLEM 1 · A train starts from rest and accelerates at 0.50 m/s² until it reaches 80 m/s. How long does it take?

1 READ

v_0 ____ · v ____ · a ____ · find ____

2 DRAW

v–t sketch

3 SOLVE

choose: $v = v_0 + at$

4 WRITE

It takes ____ s. I chose this equation because ____ .

PROBLEM 2 · A car moving 20 m/s brakes at −4 m/s². How far does it travel before stopping?

1 READ

v_0 ____ · v ____ · a ____ · no time given

2 DRAW

sketch the stop

3 SOLVE

$v^2 = v_0^2 + 2a\Delta x$

4 WRITE

The car stops in ____ m. Negative a, positive Δx because ____ .

Hardest box: □ Read □ Draw □ Solve □ Write

R5 1 2 3

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Talk it through: choosing an equation

Language goal: I can **justify** a choice.

DIRECTIONS · Read your lines aloud. Listen, then write. Switch roles each round.

ROUND 1 · 0 → 12 m/s AT 4 m/s². HOW LONG?



0 → 12 m/s
a = 4 m/s²

A ASKS "Which equation should we use?"

B SAYS "We know v0, v and a. We need ____, so I choose ____."

A CHECKS "So t = ____ s. I agree / disagree because ____ ."

ROUND 2 · SWITCH · BRAKING FROM 20 m/s AT −4 m/s²



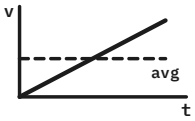
20 m/s → 0
a = −4 m/s²

B ASKS "How far until it stops? There's no time!"

A SAYS "No time means I use ____. Δx = ____ m."

B ADDS "I want to add that a is negative but Δx is ____ because ____."

ROUND 3 · BOTH · CHECK WITH AVERAGE VELOCITY



A "For steady a, average v = (v0 + v) ÷ 2 = ____."

B "Then Δx = average v × t = ____. It matches / doesn't match."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

I choose an equation by...

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

R5 1 2 3

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Which equation fits?

08 N

DIRECTIONS · Read each problem card. Write its letter under the equation you'd use. (Steady a for all.)

A Find final v after 3 s at 2 m/s².	**B** How far in 4 s from rest at 3 m/s²?	**C** Braking distance from 20 m/s at −4 m/s².	**D** How long to reach 12 m/s from 0 at 4 m/s²?
E Speed after rolling 9 m from rest at 2 m/s².	**F** Displacement in 5 s starting at 2 m/s, a = 1.	**G** Launch at 30 m/s, a = −10: velocity at 2 s?	**H** Stopping from 15 m/s in 5 m: find a.
I How far while 0 → 8 m/s in 4 s?	**J** Time to stop from 10 m/s at −2 m/s².	**K** Top of a toss: v = 0. Height if v0 = 20 m/s?	**L** Car at 5 m/s, a = 3 for 2 s: final v?

$v = v_0 + at$	$\Delta x = v_0 t + \frac{1}{2}at^2$	$v^2 = v_0^2 + 2a\Delta x$
need v or t	time is given	no time given

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

R5 1 2 3

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